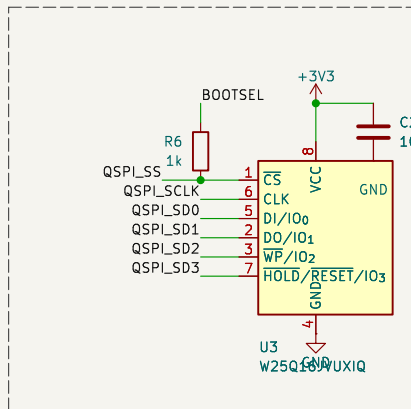


The image shows two pin headers for the Raspberry Pi 4B. The left header, labeled J2 (LEFT), has 16 pins. Pins 1, 3, 5, and 6 are connected to a +1V8 power source. Pins 2, 4, 7, 8, 9, 10, 11, 12, 13, 14, 15, and 16 are connected to a common ground (GND). The right header, labeled J3 (RIGHT), also has 16 pins. Pins 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, and 16 are connected to a common ground (GND). The pins are numbered 1 through 16 on both headers.

The RP2350A, it's core regulator, crystal, flash and decoupling are Raspberry Pi's RP2350A minimal design (RP-006440), part for part: the crystal is a 2520 with 27 p where that has an ABM8 and 15 p.

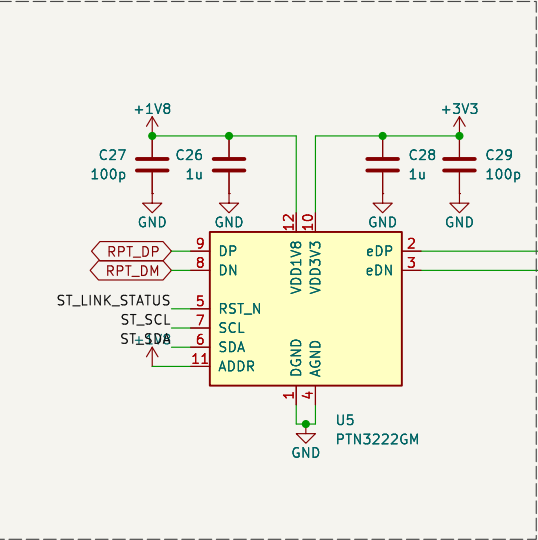
GPIO: 0-7 right-hand header, 8/9 USB data selects. 10 RF_EN, 11 1.8 V enable, 12/13 tunnelled SDA/SCL (I2C0), 14 LINK_STATUS, 15 MODE_CNTL, 16/17 tunnelled TXD/RXD (UART0), 18/19 configuration bus, 20-21, 20 translator enable, 21-24, 26-29 left-hand header, 25 LED.

The headers carry the 3.3 V side of every line the ST60 takes from a host, so a host that is not the RP2350 can run the link.

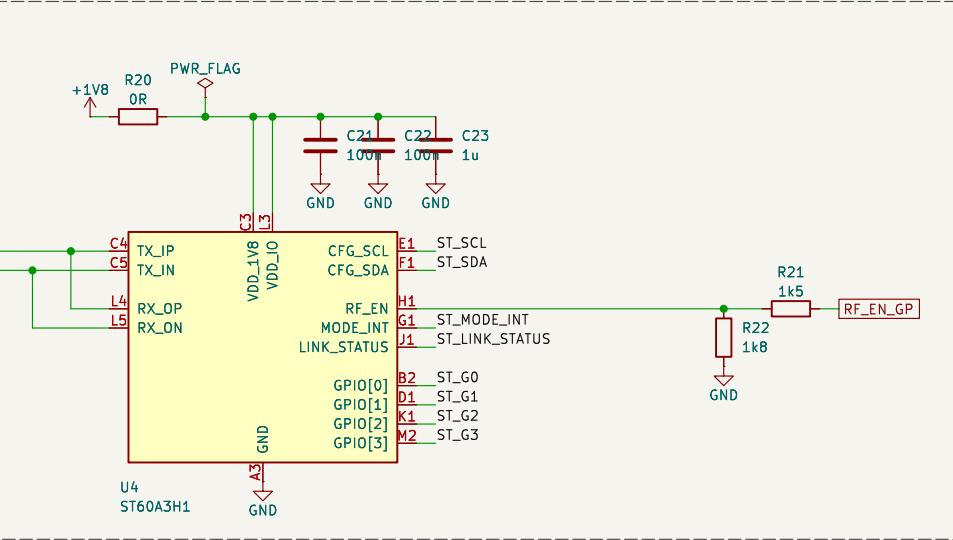


The diagram shows two switches, SW1 and SW2, connected to the BOOTSEL and RUN pins of the microcontroller. SW1 is connected to BOOTSEL and GND. SW2 is connected to RUN and GND through a 1k resistor R4.

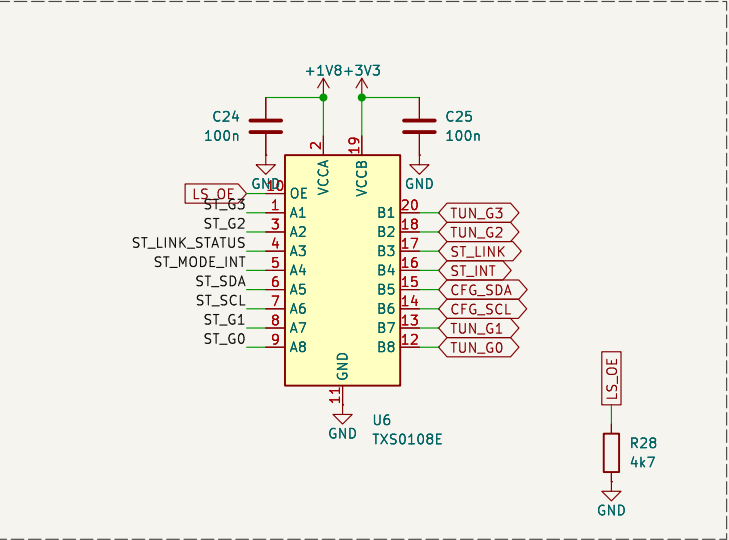
eUSB2 repeater



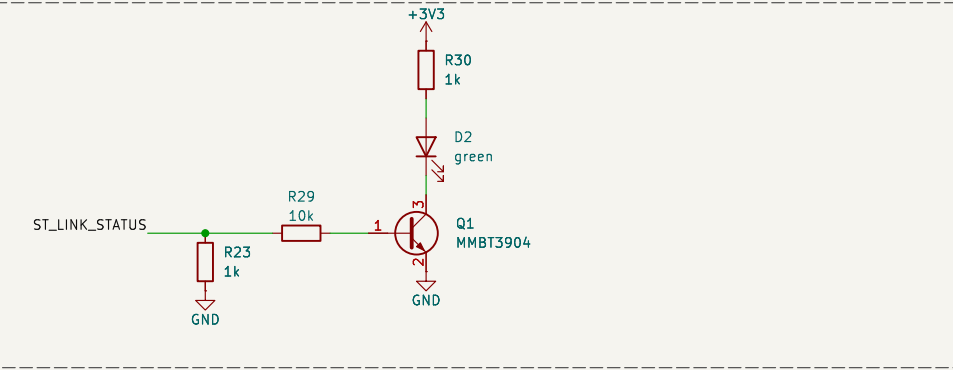
ST60A3H1



1.8 V – 3.3 V translator

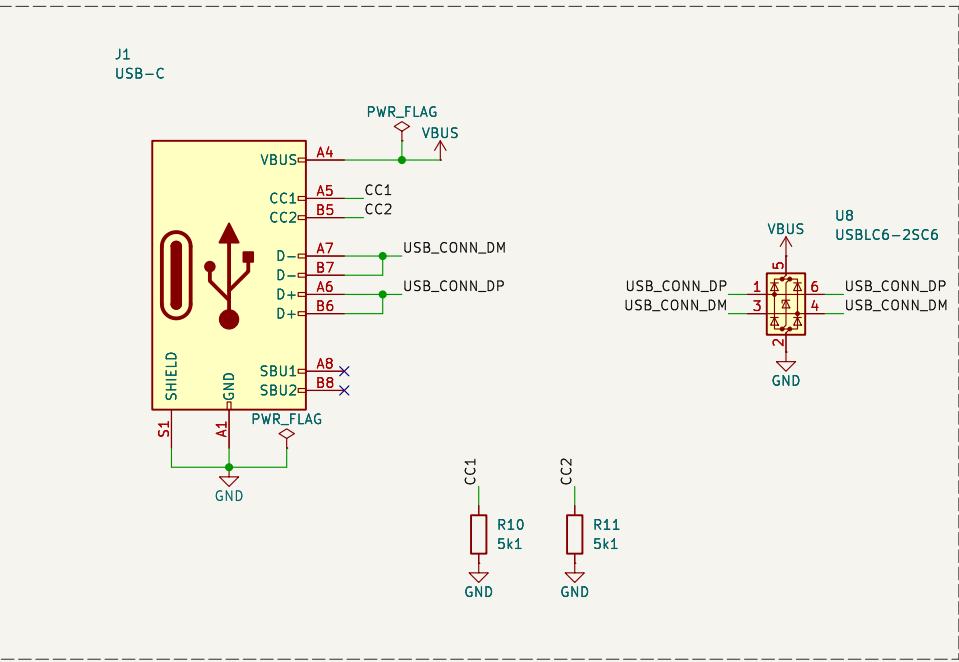


Link LED

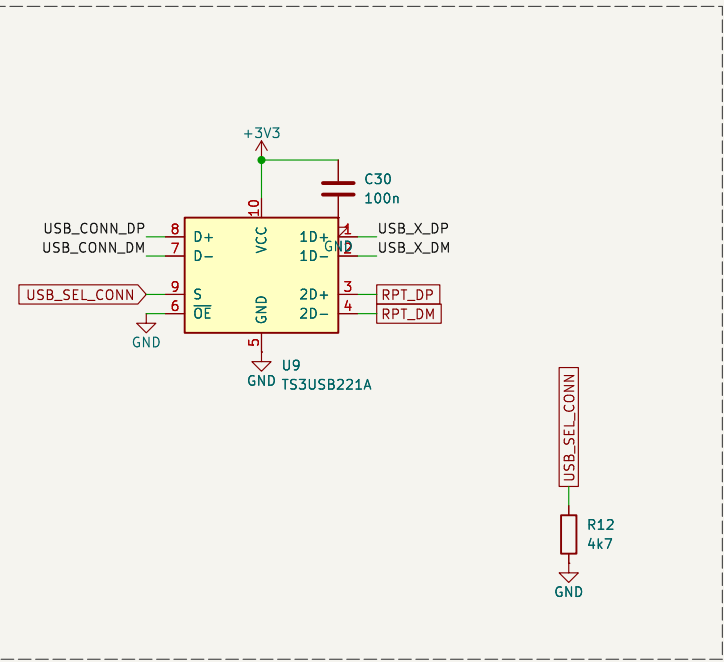


After ST's X-NUCLEO-60K1A1 (B2379A): the ST60A3H1 on +1V8 through a 0 R link, the PTN3222 on its eUSB2 pair and held in reset by LINK_STATUS, an eight-channel auto-direction translator to 3.3 V (TXS0108E here, ST2378E there), RF_EN by divider, the link LED off an NPN.
The translator is off (LS_OE low) until the RP2350 enables it, so the ST60 powers up with only its own straps on its pins. No pull-ups are fitted on the configuration bus or the tunnel pins: ST leaves its own unfitted. A tunnelled I2C bus wants pull-ups on J2/J3, at 3.3 V.
Configuration bus: ST60A3H1 at 0x60, PTN3222 at 0x43 (ADDR on +1V8).

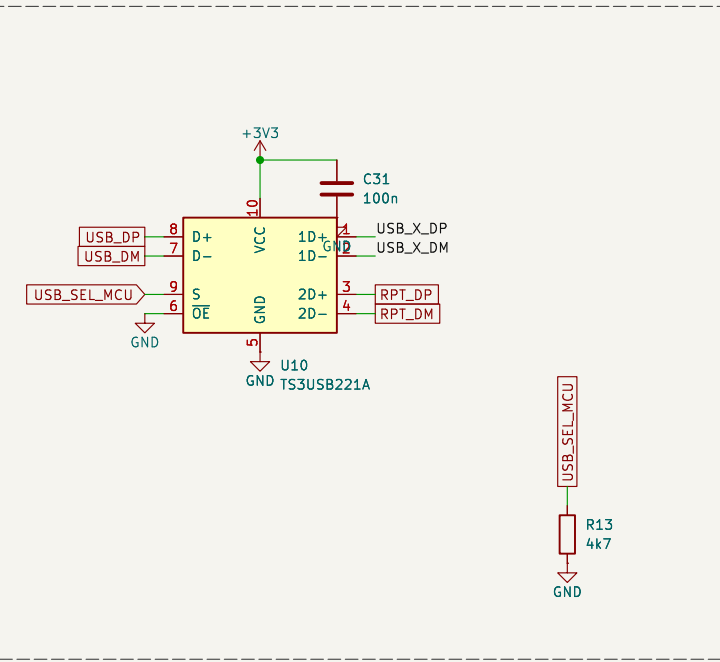
USB-C, device



Switch: connector side

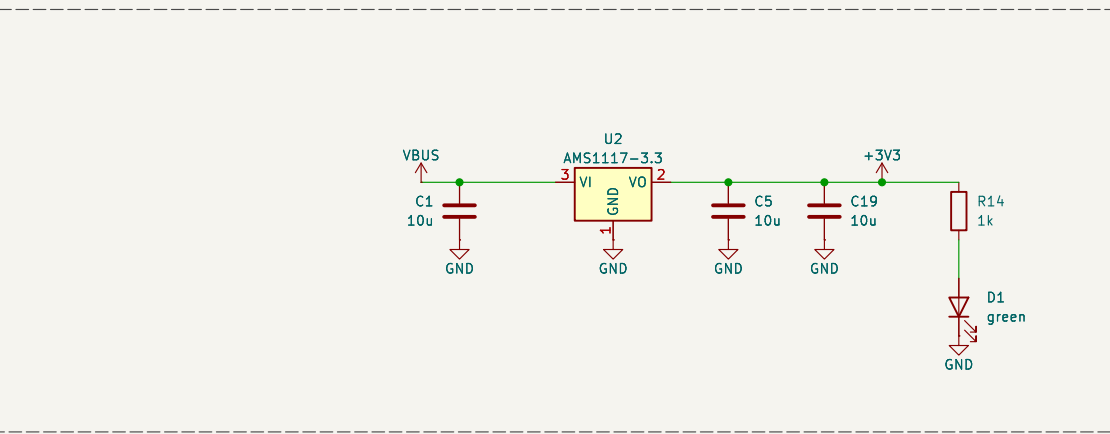


Switch: RP2350 side



Three things want the USB pair: the connector, the RP2350, the eUSB2 repeater. The two switches join any two through the node USB_X.
USB_SEL_CONN low, USB_SEL_MCU low; connector to RP2350 -- the board as it powers up, both selects pulled down.
CONN high; connector to repeater -- a host's USB goes over the air.
MCU high; RP2350 to repeater -- the RP2350 is the device at the far end.
Both high joins all three and is not a state to use.
ST_PWR_EN low turns the 1.8 V regulator off; it discharges its output, which is a power-on reset for the ST60A3H1.

3.3 V



1.8 V, switchable

